

Assignment-01

Course Code: CSE 2205

Course Title: Theory of Computing

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(Assignment - 01)

* NP: (Nondeterministic Polynomial Time)

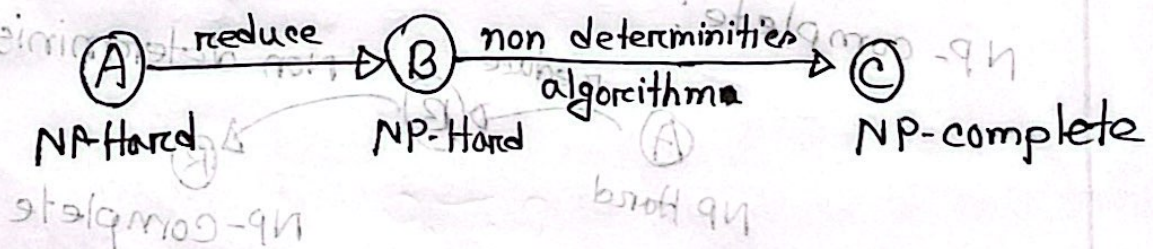
The
A problem which can not be solved in polynomial time but can be verified by polynomial time.

Example: Subset sum problem [Given a set of integers and target sum.]

2	7	1	4
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E S L

* NP-complete: A subset of NP problems that are the hardest problem in NP. A problem P is NP-complete if P is in NP and every problem in NP can be reduced to P in polynomial time.



Example:

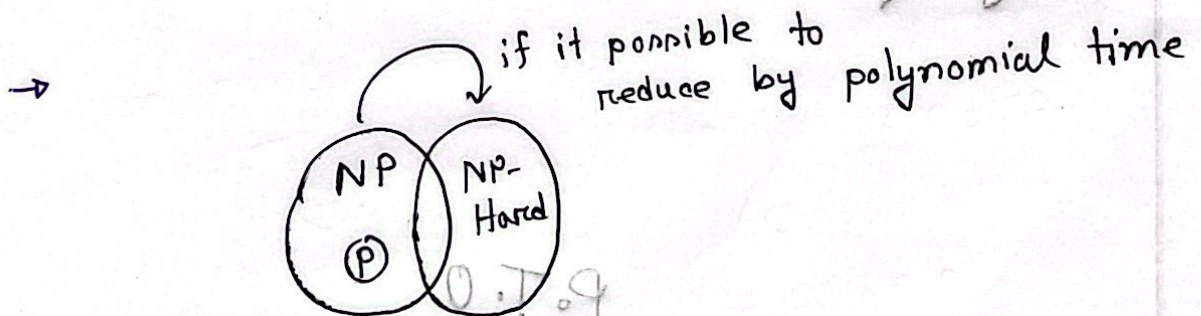
Traveling Salesman Problem: Given a list of

cities & distances between them, is there any route visiting each city once with a total distance less than a given value.

* NP-Hard: A class of problems at least as hard as the hardest problem in NP. NP-hard problems do not need to be in NP. Solving an NP-hard problem in polynomial time would solve all NP problems as well.

Example:

Halting Problem: Determine if a given program halts or run forever.)

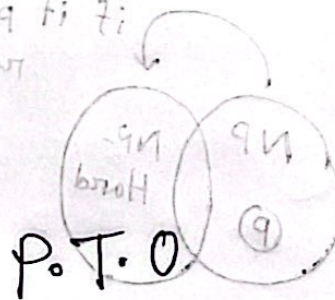


* Intractable Problems:

Inter Intractable problems are those for which solving instances of realistic size is computationally impractical. These problems have no efficient algorithm.

Example: Exponential time problems, Subset Sum Problem, undecidable problems, NPC etc.

* Different between the concept of language in automata theory & programming languages.



Language in Automata Theory	Language in Programming
① Set of strings over an Alphabet	① Formal system for writing executable programs.
② Abstract, Mathematical	② Concrete, practical
③ Not meant to be executed.	③ Designed to be compiled.
④ Theoretical study of computation & Machine models.	④ Practical use for developing & software.